

Enhancing Next Activity Prediction in Process Mining with Retrieval-Augmented Generation

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What Is Next Activity Prediction?

Predictive Process Monitoring forecasts how a running case will unfold, i.e., its outcome, remaining time, or the activities still to come.

Given a **prefix trace**, next activity prediction asks: *what event happens immediately after?*

- Take proactive measures before critical events occur
- Optimize resource allocation ahead of time
- Shift monitoring from reactive to anticipatory



Prefix trace $\langle a, b, c \rangle$ observed so far,
predict the next activity in the sequence.

- C. Di Francescomarino, I. Donadello, F. M. Maggi. Predictive Process Monitoring, Springer, 2026.
- C. Di Francescomarino, C. Ghidini. Predictive process monitoring, in: Process Mining Handbook, LNBIP, vol. 448, Springer, 2022.

The Gap: Deep Learning vs. LLMs

Deep Learning (DL): LSTM, GANs, ...

- ✓ Strong accuracy on structured event data
- ✗ Require training and feature engineering
- ✗ Limited generalization across domains

Large Language Models (LLMs)

- ✓ In-context learning
- ✗ Ignore past traces and data attributes
- ✗ Prompting alone underperforms DL



Can we ground LLMs directly in an event log via Retrieval-Augmented Generation (RAG) without further training?



RQ1

Effectiveness with/without RAG across domains

RQ2

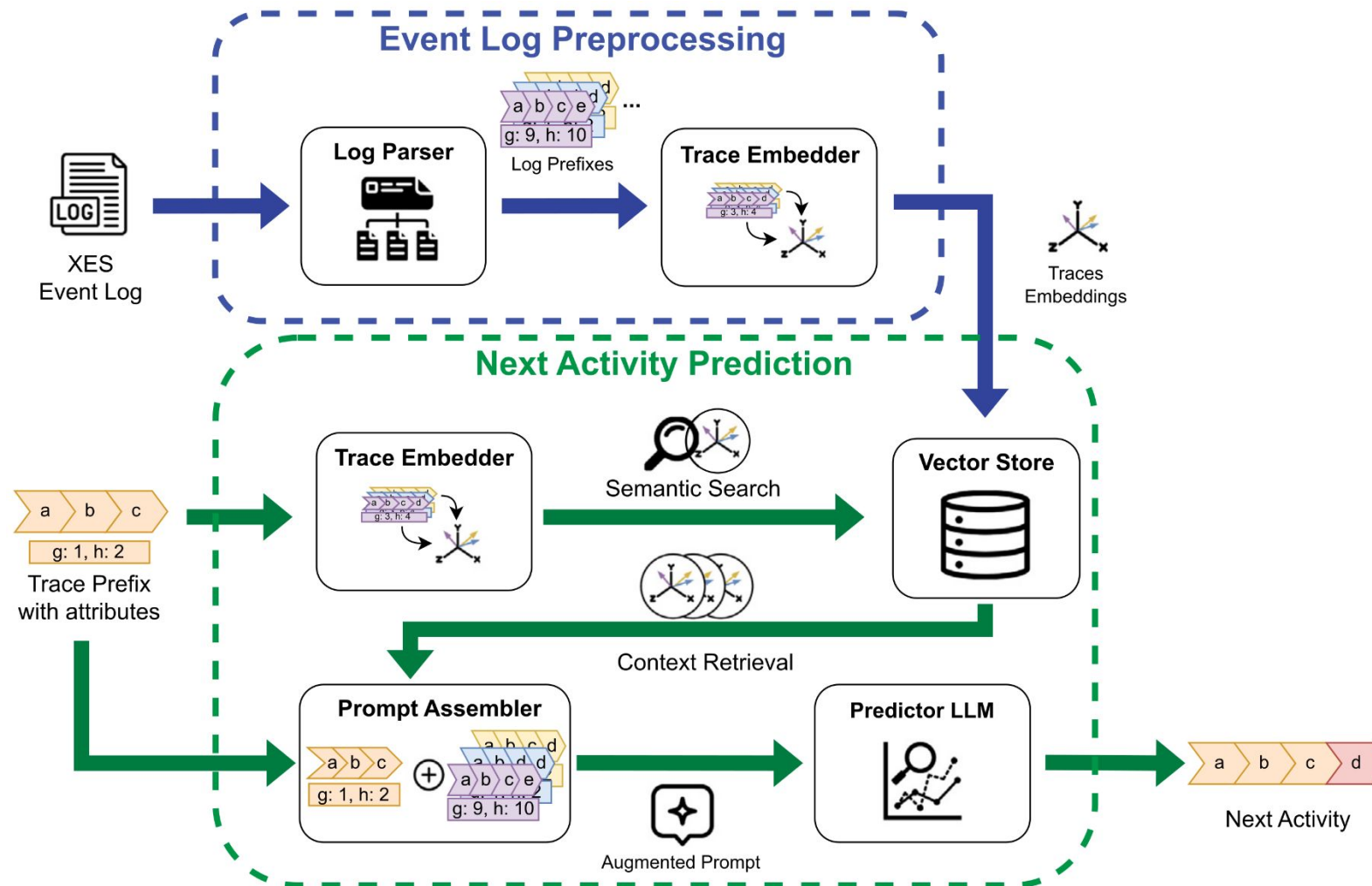
Effectiveness vs. state-of-the-art baselines

RQ3

Robustness of the employed LLMs

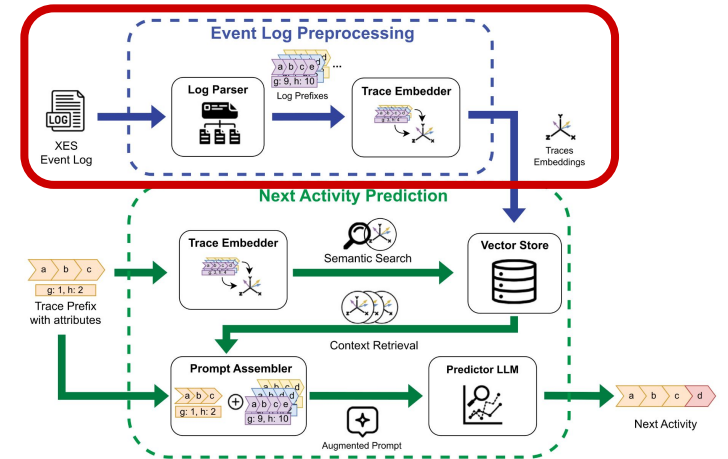
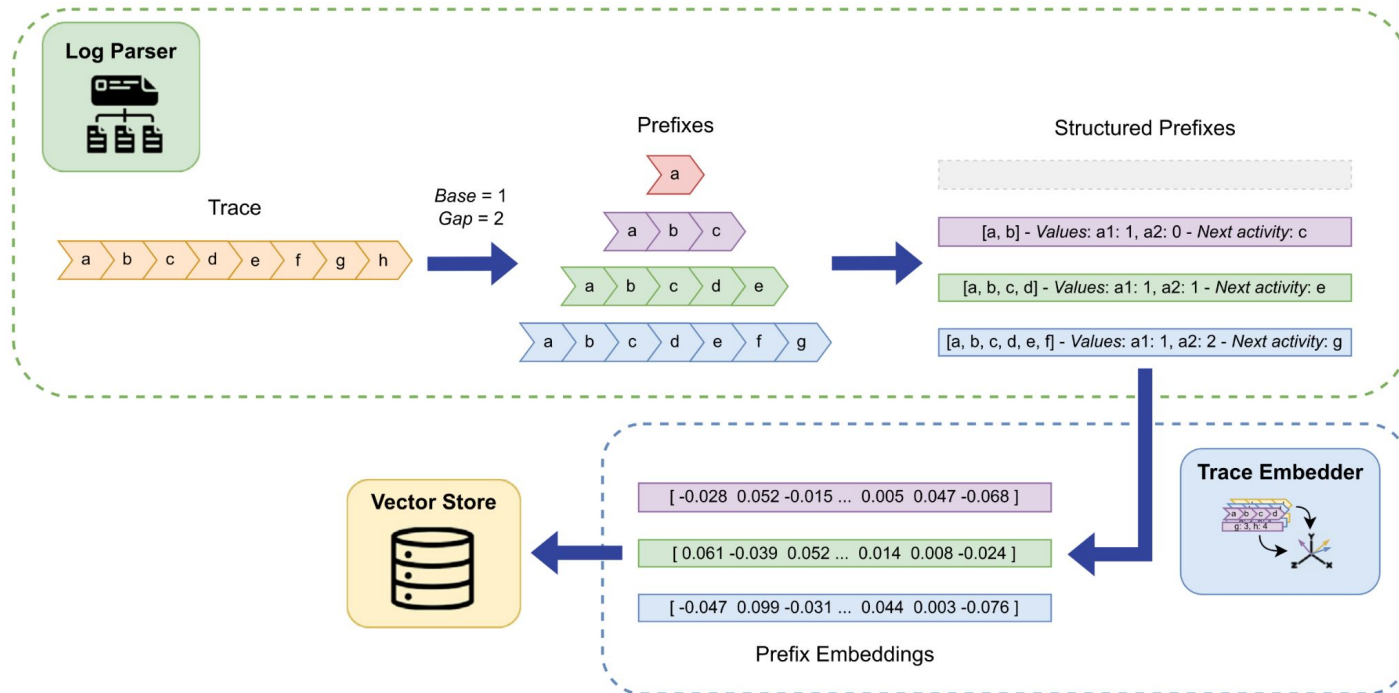
- M. Camargo et al. Learning accurate LSTM models of business processes, in: BPM 2019, Springer, 2019
- F. Taymouri et al. Predictive business process monitoring via generative adversarial nets: The case of next event prediction, 2020, CoRR abs/2003.11268
- A. Rebmam et al. Evaluating the ability of llms to solve semantics-aware process mining tasks. in: ICPM 2024, IEEE, 2024.

Retrieval-Augmented Framework



Both phases share the same vector index: prefixes stored offline are retrieved online to enrich the prompt.

Offline: From Event Log to Vector Store



Why a gap-based strategy?

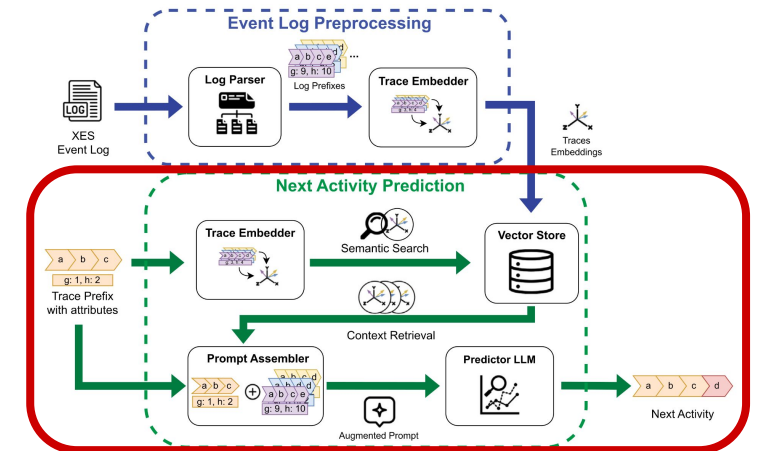
- Storing every prefix biases retrieval toward the longest traces and inflates the index.
- Sampling at fixed intervals keeps storage and retrieval tractable.

Why only final attribute values?

- Keep context window manageable.
- Preserve semantic relevance for retrieval.
- Avoid the “lost-in-the-middle” effect.
- N.F. Liu et al. Lost in the middle: How language models use long contexts, Trans. Assoc. Comput. Linguist. 12 (2024) 157–173

Online: Retrieval-Augmented Prediction

1. **Embed** the incoming trace prefix
2. **Cosine-similarity search** over the vector store
3. **Retrieve** the **top- k** most similar past prefixes
4. Assemble the retrieved traces into an **augmented prompt**
5. The Predictor LLM outputs the **next activity**



$k = 3$

Provide enough context &
avoid lost-in-the-middle
effect

- N.F. Liu et al. Lost in the middle: How language models use long contexts, Trans. Assoc. Comput. Linguist. 12 (2024) 157–173

Experimental Setup



9 Event Logs

Real-world

- BPIC 2012
- BPIC 2013 (Closed/Incidents)
- BPIC 2020 (Int. Decl.)
- Hospital Billing
- Sepsis Cases

Synthetic

- SL2-2v-1r
- SL3-multivariant
- Udon-ya



10 Language models

Open-weight

- Llama 3.1 8B, 3.2 (1B, 3B)
- Mistral 7B (v0.2, v0.3)
- Qwen 2.5 7B Instruct
- Phi 4 (14B)
- DeepSeek-R1 Distilled (Qwen-7B, Llama-8B)

Proprietary

- GPT 4o Mini

+ 6 Embedding models



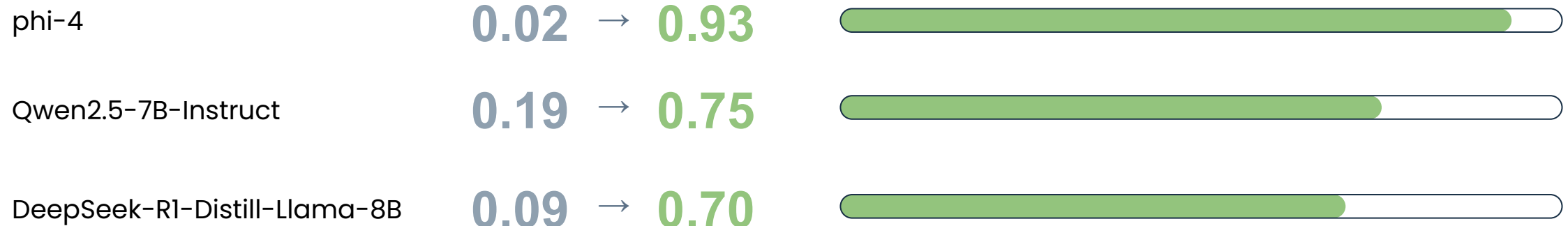
Protocol

- 70%/30% retrieval-test split
- 300 test prefixes per event log
- *Top-3* retrieval, *single-bucket* default
- Metrics:
 - *accuracy*,
 - *macro precision*, *recall*, *F1*
- Earliness, robustness, and statistical significance

RQ1: With vs. Without RAG

RAG Boosts Accuracy of Standalone LLMs

Accuracy on BPI Challenge 2020 (International Declarations)



The same pattern holds for every model and event log we tested: **similar past traces** are what make LLMs **usable** for this task.

RQ1: Best-Performing LLM per Event Log

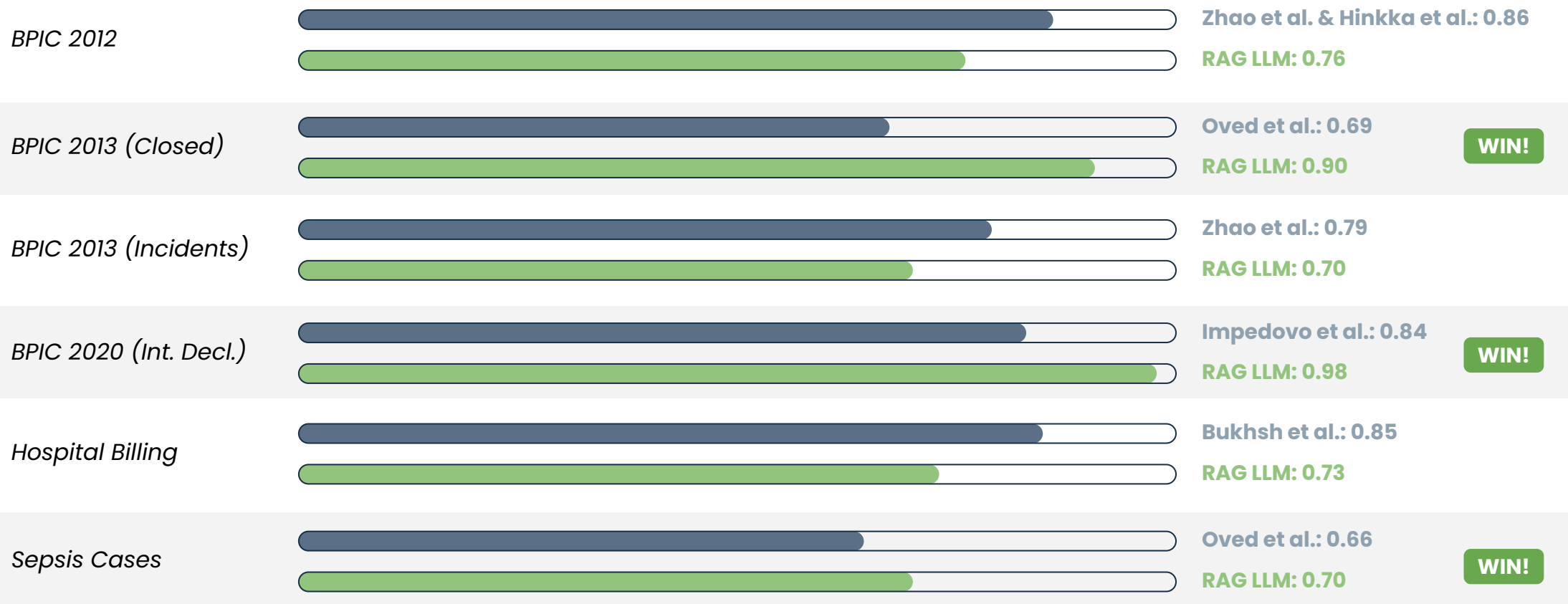
Event Log	Type	Best Model	Accuracy	Avg. Time Per-Inference (s)
<i>BPIC 2012</i>	Real	gpt-4o-mini	0.76	20.1
<i>BPIC 2013 (Closed Problems)</i>	Real	gpt-4o-mini	0.90	13.6
<i>BPIC 2013 (Incidents)</i>	Real	gpt-4o-mini	0.70	21.5
<i>BPIC 2020 (Int. Decl.)</i>	Real	gpt-4o-mini	0.98	11.0
<i>Hospital Billing</i>	Real	gpt-4o-mini	0.73	14.2
<i>Sepsis Cases</i>	Real	phi-4	0.70	41.6
<i>SL2-2v-1r</i>	Synthetic	Qwen2.5-7B-Instruct	0.81	30.5
<i>SL3-multivariant</i>	Synthetic	gpt-4o-mini	0.47	12.4
<i>Udon-ya</i>	Synthetic	gpt-4o-mini	0.84	16.4

phi-4 and Qwen2.5-7B-Instruct are the strongest open-weight alternatives. Llama 3.2 variants falls short on every log.

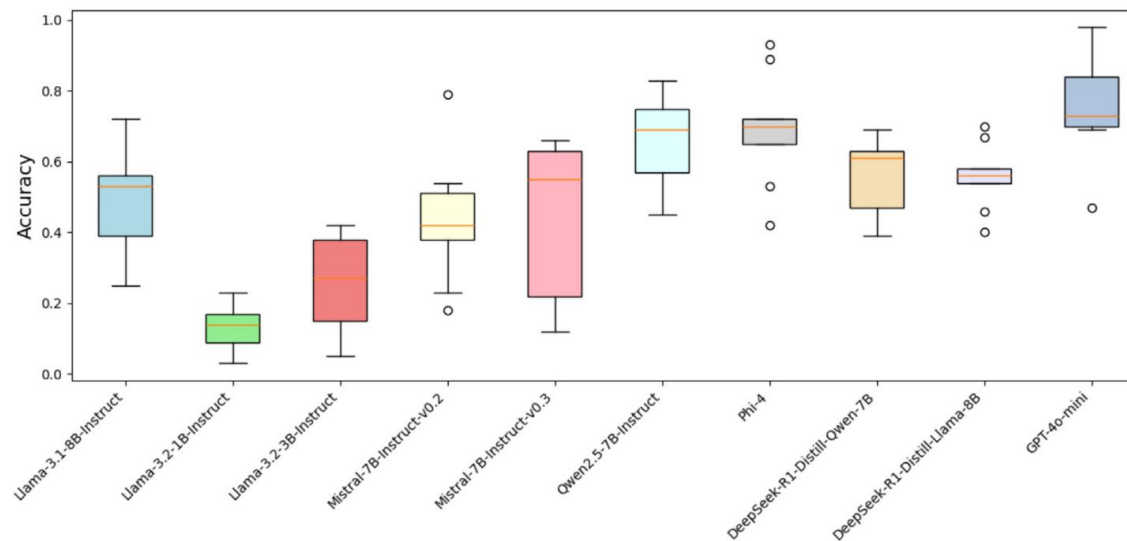
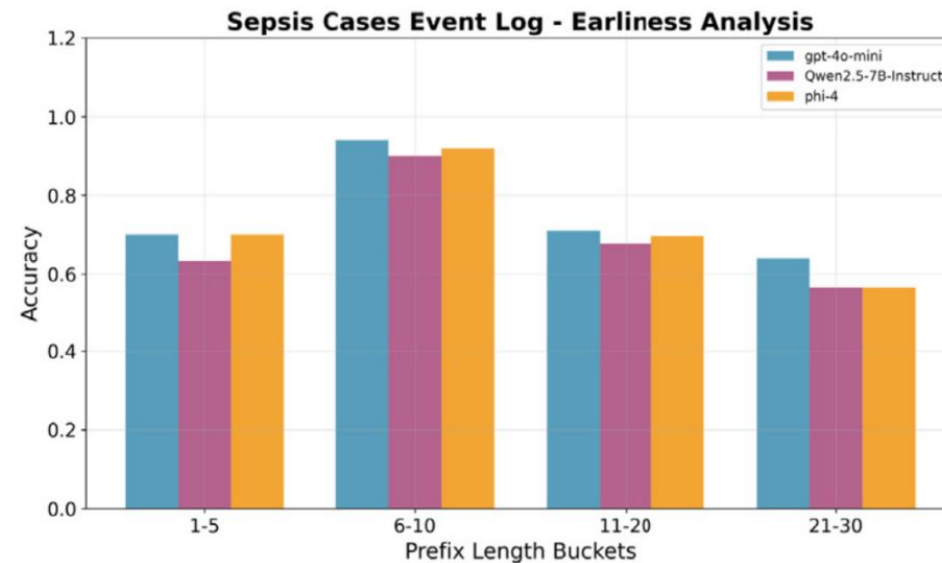
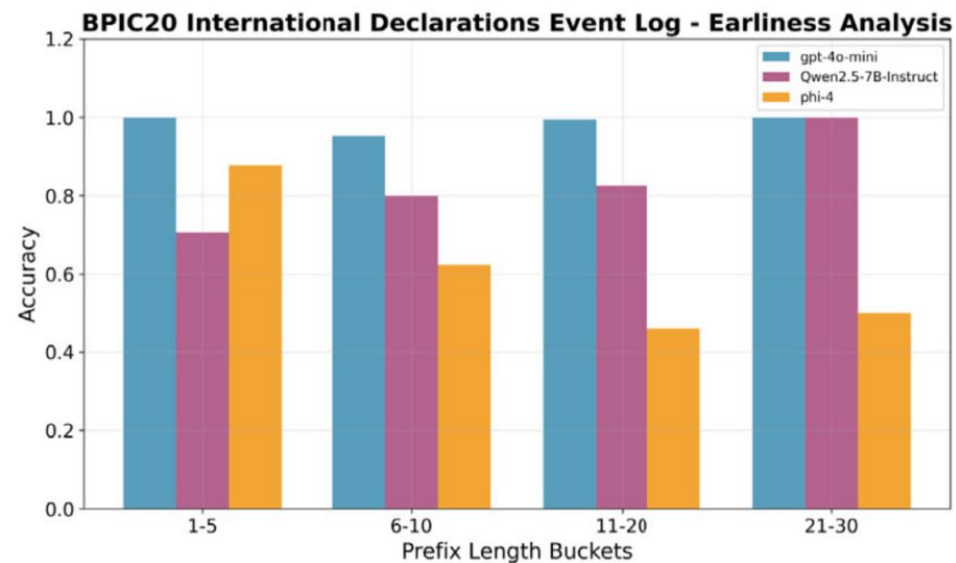
RQ2: State of the Art Comparison

Competitive Without Any Fine-Tuning

Accuracy vs. the best reported literature baseline, per real-world log



RQ3: Earliness and Robustness



Limitations and Threats to Validity



Limitations

- Accuracy drops on logs with frequent interleavings and full lifecycle stages
- Concept drift is not considered
- Phi-4 LoRA fine-tuning on S_NAP degraded accuracy (0.93→0.39 on BPIC 2020), despite faster inference



Threats to Validity

- *Rapid LLM evolution*: mitigated by benchmarking 10 models
- *Stochastic LLM outputs*: published prompts and repeated runs
- *Generalizability*: addressed with 9 logs across business domains

- A. Rebmann et al. Evaluating the ability of llms to solve semantics-aware process mining tasks. in: ICPM 2024, IEEE, 2024.

CONCLUSION



Contributions

- RAG turns LLMs into effective next-activity predictors without fine-tuning
- Competitive with trained DL baselines
- Findings are statistically significant



Future Work

- Handling concept drift in evolving process behaviors
- Standardized benchmarks for LLM-based PPM
- Efficient RAG architectures (e.g. FastRAG) for reduced overhead



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THANK YOU

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Journal

Paper



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About Me



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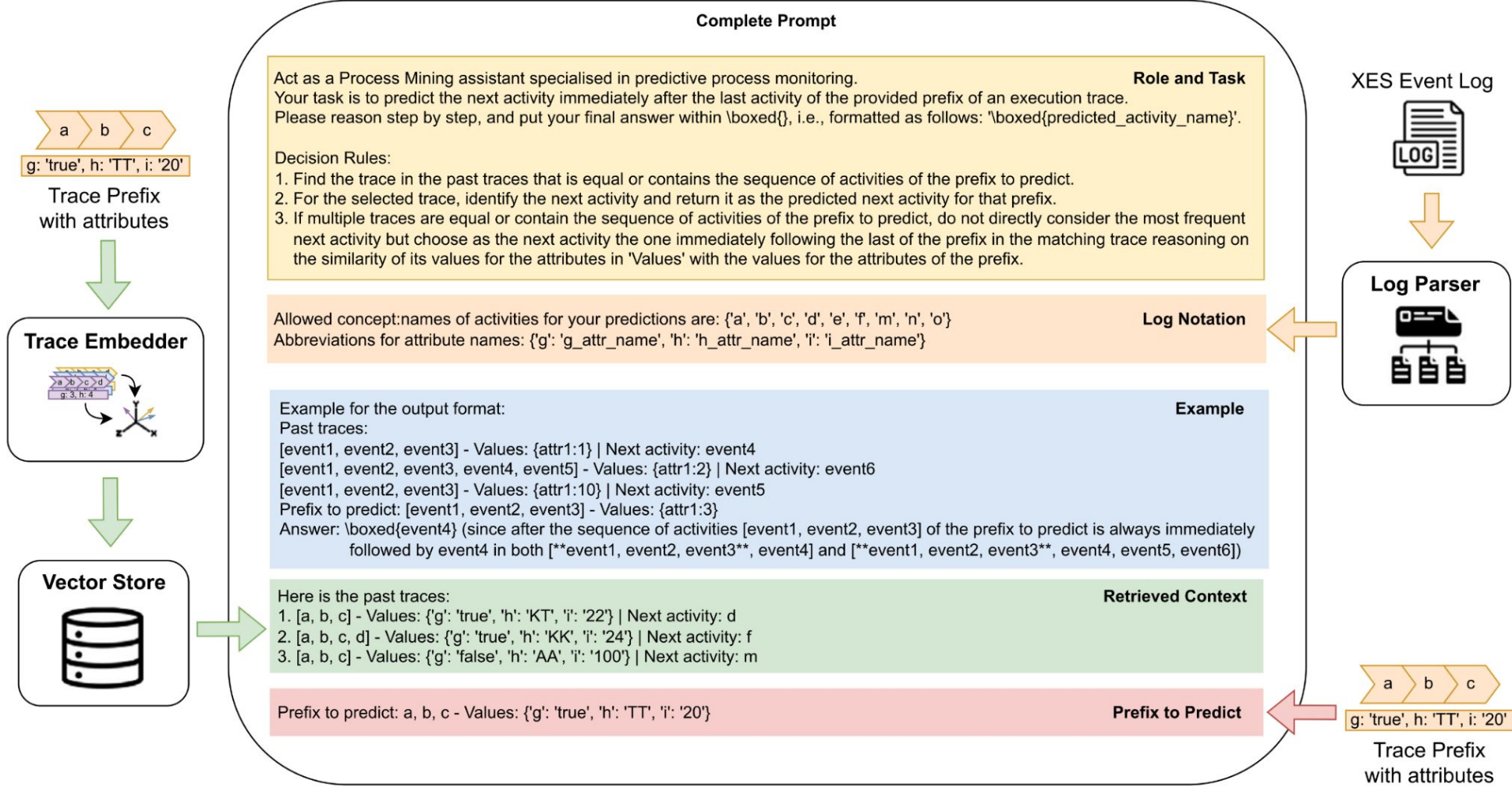


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Prompt Design for Reliable Predictions



RQ1: With vs. Without RAG

Language model	Event Log	RAG	Accuracy
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>		0.19
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	✓	0.75
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>		0.02
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	✓	0.93
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>		0.09
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	✓	0.70
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>		0.36
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>	✓	0.46
<i>phi-4</i>	<i>SL3-multivariant</i>		0.06
<i>phi-4</i>	<i>SL3-multivariant</i>	✓	0.42
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>		0.09
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>	✓	0.40

RQ1: Bucketing Strategies

Language model	Event log	Gap	Accuracy
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	1	0.75
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	1	0.93
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	1	0.70
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	3	0.81
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	3	0.94
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	3	0.64
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	5	0.85
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	5	0.91
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	5	0.70
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>	1	0.46
<i>phi-4</i>	<i>SL3-multivariant</i>	1	0.42
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>	1	0.40
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>	3	0.46
<i>phi-4</i>	<i>SL3-multivariant</i>	3	0.44
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>	3	0.40
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>	5	0.46
<i>phi-4</i>	<i>SL3-multivariant</i>	5	0.44
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>	5	0.41

RQ1: Embedding Models

Language model	Event Log	Embedding model	Accuracy
Qwen2.5-7B-Instruct	BPI Challenge 2020 (Int. Decl.)	all-MiniLM-L6-v2	0.80
phi-4	BPI Challenge 2020 (Int. Decl.)	all-MiniLM-L6-v2	0.92
Qwen2.5-7B-Instruct	BPI Challenge 2020 (Int. Decl.)	all-MiniLM-L12-v2	0.75
phi-4	BPI Challenge 2020 (Int. Decl.)	all-MiniLM-L12-v2	0.93
Qwen2.5-7B-Instruct	BPI Challenge 2020 (Int. Decl.)	all-mpnet-base-v2	0.76
phi-4	BPI Challenge 2020 (Int. Decl.)	all-mpnet-base-v2	0.90
Qwen2.5-7B-Instruct	BPI Challenge 2020 (Int. Decl.)	jina-embeddings-v2-base-en	0.75
phi-4	BPI Challenge 2020 (Int. Decl.)	jina-embeddings-v2-base-en	0.88
Qwen2.5-7B-Instruct	BPI Challenge 2020 (Int. Decl.)	jina-embeddings-v3	0.54
phi-4	BPI Challenge 2020 (Int. Decl.)	jina-embeddings-v3	0.55
Qwen2.5-7B-Instruct	BPI Challenge 2020 (Int. Decl.)	jina-embeddings-v4	0.53
phi-4	BPI Challenge 2020 (Int. Decl.)	jina-embeddings-v4	0.56
Qwen2.5-7B-Instruct	SL3-multivariant	all-MiniLM-L6-v2	0.45
phi-4	SL3-multivariant	all-MiniLM-L6-v2	0.44
Qwen2.5-7B-Instruct	SL3-multivariant	all-MiniLM-L12-v2	0.46
phi-4	SL3-multivariant	all-MiniLM-L12-v2	0.42
Qwen2.5-7B-Instruct	SL3-multivariant	all-mpnet-base-v2	0.46
phi-4	SL3-multivariant	all-mpnet-base-v2	0.46
Qwen2.5-7B-Instruct	SL3-multivariant	jina-embeddings-v2-base-en	0.46
phi-4	SL3-multivariant	jina-embeddings-v2-base-en	0.45
Qwen2.5-7B-Instruct	SL3-multivariant	jina-embeddings-v3	0.21
phi-4	SL3-multivariant	jina-embeddings-v3	0.27
Qwen2.5-7B-Instruct	SL3-multivariant	jina-embeddings-v4	0.17
phi-4	SL3-multivariant	jina-embeddings-v4	0.29

RQ1: With vs. Without RAG

Language model	Event Log	RAG	Accuracy
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>		0.19
<i>Qwen2.5-7B-Instruct</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	✓	0.75
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>		0.02
<i>phi-4</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	✓	0.93
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>		0.09
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>BPI Challenge 2020 (Int. Decl.)</i>	✓	0.70
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>		0.36
<i>Qwen2.5-7B-Instruct</i>	<i>SL3-multivariant</i>	✓	0.46
<i>phi-4</i>	<i>SL3-multivariant</i>		0.06
<i>phi-4</i>	<i>SL3-multivariant</i>	✓	0.42
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>		0.09
<i>DeepSeek-R1-Distill-Llama-8B</i>	<i>SL3-multivariant</i>	✓	0.40

RQ1: Performance VS. Time

Table 6

Accuracy, (macro) precision, (macro) recall, and (macro) F1-score in the next activity prediction task for the *BPI Challenge 2013 (Incidents)* log. The **best** and *second-best* results for each metric are reported in **bold** and *italics*, respectively. The execution time is reported as total time (in *hours*) and average per-inference time (in *seconds*).

Language model	Accuracy	Precision	Recall	F1-score	Time	
					Total (h)	Per-Inference (s)
<i>Llama-3.1-8B-Instruct</i>	0.37	0.33	0.32	0.32	5.28	63.3
<i>Llama-3.2-1B-Instruct</i>	0.03	0.03	0.02	0.03	2.65	31.8
<i>Llama-3.2-3B-Instruct</i>	0.05	0.06	0.05	0.05	3.05	36.5
<i>Mistral-7B-Instruct-v0.2</i>	0.38	0.31	0.30	0.30	4.49	53.8
<i>Mistral-7B-Instruct-v0.3</i>	0.12	0.12	0.11	0.11	4.38	52.5
<i>Qwen2.5-7B-Instruct</i>	0.45	0.32	0.32	0.31	2.71	32.5
<i>phi-4</i>	0.53	0.57	0.53	0.54	5.15	61.7
<i>DeepSeek-R1-Distill-Qwen-7B</i>	0.39	0.38	0.37	0.37	5.69	68.2
<i>DeepSeek-R1-Distill-Llama-8B</i>	0.46	0.43	0.42	0.43	9.52	114.2
<i>gpt-4o-mini</i>	0.70	0.64	0.65	0.64	1.79	21.5

Table 7

Accuracy, (macro) precision, (macro) recall, and (macro) F1-score in the next activity prediction task for the *BPI Challenge 2020 (International Declarations)* log. The **best** and *second-best* results for each metric are reported in **bold** and *italics*, respectively. The execution time is reported as total time (in *hours*) and average per-inference time (in *seconds*).

Language model	Accuracy	Precision	Recall	F1-score	Time	
					Total (h)	Per-Inference (s)
<i>Llama-3.1-8B-Instruct</i>	0.71	0.74	0.68	0.67	2.65	31.8
<i>Llama-3.2-1B-Instruct</i>	0.23	0.31	0.18	0.19	0.71	8.5
<i>Llama-3.2-3B-Instruct</i>	0.39	0.61	0.42	0.48	2.05	24.5
<i>Mistral-7B-Instruct-v0.2</i>	0.79	0.74	0.71	0.71	2.34	28.1
<i>Mistral-7B-Instruct-v0.3</i>	0.63	0.50	0.47	0.46	1.29	15.4
<i>Qwen2.5-7B-Instruct</i>	0.75	0.82	0.80	0.78	2.13	25.5
<i>phi-4</i>	0.93	0.94	0.93	0.93	3.12	37.3
<i>DeepSeek-R1-Distill-Qwen-7B</i>	0.69	0.59	0.58	0.55	3.93	47.2
<i>DeepSeek-R1-Distill-Llama-8B</i>	0.70	0.70	0.69	0.67	7.34	88.1
<i>gpt-4o-mini</i>	0.98	0.99	0.99	0.99	0.92	11.0

RQ2: State of the Art Comparison Complete

Dataset	Study	Accuracy
<i>BPI Challenge 2012</i>	[3]	0.85
	[36]	0.59
	[40]	0.83
	[42]	0.83
	[69]	0.86
	[70]	0.78
	[71]	0.86
	[41]	0.84
	[72]	0.42
	[35] (w/o attributes)	0.82
	[35] (w/ attributes)	0.80
	[73]	0.85
	Our approach	0.76
<i>BPI Challenge 2013 (Closed Problems)</i>	[3]	0.64
	[11]	0.69
	[36]	0.58
	[40]	0.24
	[42]	0.54
	[71]	0.63
	[41]	0.24
	[72]	0.43
	[35] (w/o attributes)	0.59
	[35] (w/ attributes)	0.54
	Our approach	0.90
<i>BPI Challenge 2013 (Incidents)</i>	[3]	0.70
	[11]	0.67
	[36]	0.66
	[40]	0.46
	[42]	0.66
	[69]	0.79
	[71]	0.74
	[41]	0.36
	[72]	0.51
	[35] (w/o attributes)	0.59
	[35] (w/ attributes)	0.51
	[73]	0.62
	Our approach	0.70

<i>BPI Challenge 2020 (Int. Decl.)</i>	[74]	0.84
	Our approach	0.98
<i>Hospital Billing</i>	[3]	0.78
	[40]	0.75
	[69]	0.84
	[71]	0.79
	[72]	0.70
	[73]	0.85
	Our approach	0.73
<i>Sepsis Cases</i>	[3]	0.64
	[11]	0.66
	[36]	0.40
	[40]	0.56
	[69]	0.62
	[70]	0.65
	[71]	0.63
	[72]	0.21
	[35] (w/o attributes)	0.55
	[35] (w/ attributes)	0.58
	[75]	0.60
	Our approach	0.70

RQ3: Statistical Significance

Nemenyi Test

- GPT 4o Mini confirmed to be the best.
- Phi-4 and Qwen 2.5 7B are ahead of other smaller open-weight models ($p < 0.05$).

Friedman Test

$$\chi^2 = 25.56$$

$$p = 1.18 \times 10^{-5}$$

Model rankings differ significantly,
not by chance

RQ3: Nemenyi & Fine-Tuning

Table 14

Extract of Nemenyi test results for the three best models, indicating the p -values comparing each pair of LLMs (row vs. column). Lower values report stronger evidence against the null hypothesis H_0 of equal performance.

	<i>gpt-4o-mini</i>	<i>phi-4</i>	<i>Qwen2.5-7B-Instruct</i>
<i>gpt-4o-mini</i>	1.000	0.726	0.002
<i>phi-4</i>	0.726	1.000	0.046
<i>Qwen2.5-7B-Instruct</i>	0.002	0.046	1.000

Table 15

Accuracy, (macro) precision, (macro) recall, and (macro) F1-score achieved by the framework with the fine-tuned *phi-4* model in the next activity prediction task across all the considered event logs. The execution time is reported as total time (in *hours*) and average per-inference time (in *seconds*).

Event Log	Accuracy	Precision	Recall	F1-score	Time	
					Total (h)	Per-Inference (s)
<i>BPIC 2012</i>	0.37	0.18	0.26	0.19	2.06	24.8
<i>BPIC 2013 (Closed Problems)</i>	0.11	0.10	0.10	0.10	0.69	8.2
<i>BPIC 2013 (Incidents)</i>	0.27	0.29	0.27	0.27	2.06	24.7
<i>BPIC 2020 (Int. Decl.)</i>	0.39	0.35	0.38	0.34	1.64	19.6
<i>Hospital Billing</i>	0.44	0.49	0.39	0.42	1.53	18.3
<i>Sepsis Cases</i>	0.39	0.28	0.28	0.27	1.85	22.2
<i>SL2-2v-1r</i>	0.35	0.16	0.10	0.11	0.25	19.7
<i>SL3-multivariant</i>	0.49	0.13	0.08	0.10	2.16	25.8
<i>Udon-ya</i>	0.24	0.29	0.28	0.28	2.47	29.6

RQ3: Phi-4 Fine-Tuning

PEFT via LoRA



- AdamW algorithm:
 - initial learning rate of $1 \cdot 10^{-5}$
 - gradient accumulation over 16 batches (to simulate batch of 32)
- Avg training time per epoch: 21.7 h

Accuracy, (macro) precision, (macro) recall, and (macro) F1-score achieved by the framework with the fine-tuned phi-4 model in the next activity prediction task across all the considered event logs. The execution time is reported as total time (in *hours*) and average per-inference time (in *seconds*).

Event Log	Accuracy	Precision	Recall	F1-score	Time	
					Total (h)	Per-Inference (s)
BPIC 2012	0.37	0.18	0.26	0.19	2.06	24.8
BPIC 2013 (Closed Problems)	0.11	0.10	0.10	0.10	0.69	8.2
BPIC 2013 (Incidents)	0.27	0.29	0.27	0.27	2.06	24.7
BPIC 2020 (Int. Decl.)	0.39	0.35	0.38	0.34	1.64	19.6
Hospital Billing	0.44	0.49	0.39	0.42	1.53	18.3
Sepsis Cases	0.39	0.28	0.28	0.27	1.85	22.2
SL2-2v-1r	0.35	0.16	0.10	0.11	0.25	19.7
SL3-multivariant	0.49	0.13	0.08	0.10	2.16	25.8
Udon-ya	0.24	0.29	0.28	0.28	2.47	29.6